INCIDENT POST-MORTEM · INC-2026-0312

Global API Gateway *Outage*

A cascading failure in the API gateway layer caused a 4-hour service disruption affecting 2.3 million users across three regions. This report examines what happened, why, and how we prevent recurrence.

Severity SEV-1 (Critical) **Date** March 8, 2026 **Duration** 4h 23m

Prepared by Site Reliability Engineering

4h 23m	2.3M	$1.7M	99.91%
DURATION	USERS AFFECTED	REVENUE IMPACT	MARCH SLA
Total time from first alert to full recovery	Unique users who experienced errors or degradation	Estimated lost transactions during the outage window	Below the 99.95% target after this incident

EXECUTIVE SUMMARY

On March 8, 2026, a **misconfigured rate-limiting policy** deployed to the API gateway cluster triggered a cascading failure that propagated across all three production regions. The incident began at 02:17 UTC when an automated configuration push inadvertently set the per-node connection limit to 100 — two orders of magnitude below the intended 10,000 — causing immediate request queuing, upstream health check failures, and a complete loss of API availability.

The root cause was a YAML indentation error in the gateway configuration template, combined with insufficient validation in the deployment pipeline. The error passed through code review, staging validation, and canary deployment without detection because the staging environment's traffic volume was too low to trigger the misconfigured threshold.

This report provides a complete timeline of the incident, analysis of contributing factors, impact assessment, and a prioritized set of remediation actions. Three of the five identified action items have already been completed.

SECTION 1

Incident timeline

The incident progressed through four phases: trigger, escalation, mitigation, and recovery. Total elapsed time from first alert to full resolution was 4 hours 23 minutes.

EXHIBIT 1

Incident timeline

Key events from trigger to full recovery, March 8 2026 (all times UTC)

02:12 UTC

Configuration push initiated

Automated deployment pipeline pushes gateway config v3.14.2 to the `us-east-1` cluster. The change was intended to update TLS cipher suites.

02:17 UTC

First alerts fire — P99 latency spike

PagerDuty alert triggers for API p99 latency exceeding 5s threshold. Initial triage classifies as transient. On-call SRE acknowledges but does not escalate.

02:24 UTC

Cascading failure begins

Connection pool exhaustion on `us-east-1` gateway nodes causes health check failures. Load balancer begins rerouting traffic to `eu-west-1` and `ap-southeast-1`, which are still running the old config.

02:31 UTC

Global propagation — all regions affected

Configuration sync propagates v3.14.2 to remaining regions. All three clusters now experiencing connection pool exhaustion. Error rate reaches 94%. Incident escalated to SEV-1.

02:48 UTC

Root cause identified

SRE team identifies the rate-limit misconfiguration via `diff` between v3.14.1 and v3.14.2. The `max_connections_per_node` value was set to 100 instead of 10000 due to a YAML indentation error.

03:15 UTC

Rollback initiated

Decision made to roll back to v3.14.1. Rollback delayed by 27 minutes because the deployment pipeline required approval from two engineers, one of whom was unreachable.

04:42 UTC

Rollback complete — `us-east-1` recovering

First region fully restored. Stale connection pools draining. Error rate drops below 10% in `us-east-1`.

06:40 UTC

Full recovery confirmed

All three regions reporting nominal error rates (<0.1%). Incident formally closed. Total duration: 4h 23m.

Source: PagerDuty incident log, Datadog APM traces, deployment audit trail

SECTION 2

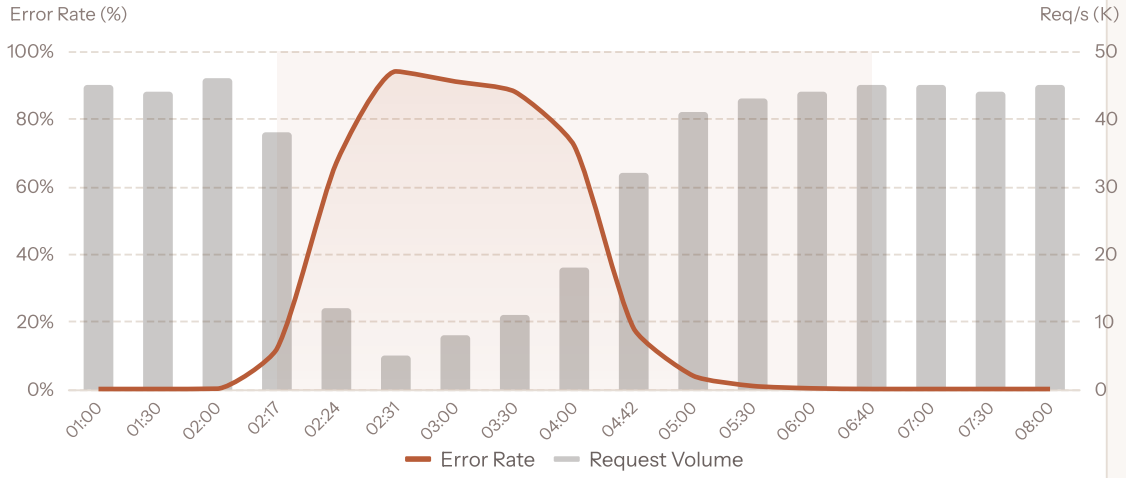
Impact analysis

The outage had significant impact across three dimensions: user experience, revenue, and downstream system integrity.

EXHIBIT 2

Error rate and request volume during incident

5-minute aggregates, March 8 2026 01:00–08:00 UTC



Source: Datadog APM, sampled at 5-minute intervals

User impact. 2.3 million unique users encountered HTTP 502/503 errors during the outage window. Mobile app users experienced the worst degradation due to aggressive retry logic that amplified the connection pool pressure. Customer support received 12,400 tickets in the first two hours — a 47× increase over normal volume.

Revenue impact. An estimated $1.7M in transactions failed to complete during the outage. Of this, $1.2M was recovered within 24 hours through automatic retry and reconciliation. Net revenue loss: approximately $500K.

Downstream systems. The payment processing queue accumulated 340,000 pending transactions during the outage. Post-recovery processing created a secondary latency spike lasting 90 minutes as the backlog cleared. Three partner API integrations triggered their own circuit breakers, requiring manual re-enablement.

SLA impact. March uptime dropped to 99.91%, below the 99.95% SLA commitment. This triggers contractual service credits for 14 enterprise customers totaling approximately $180K.

EXHIBIT 3

Impact by region

Users affected, error rate, and time to recovery by region

REGION	USERS AFFECTED	PEAK ERROR RATE	TIME TO RECOVERY	REVENUE IMPACT
us-east-1	**1,340,000**	**97.2%**	4h 23m	**$980K**
eu-west-1	620,000	89.4%	3h 51m	$460K

REGION	USERS AFFECTED	PEAK ERROR RATE	TIME TO RECOVERY	REVENUE IMPACT
ap-southeast-1	340,000	78.1%	3h 12m	$260K

Source: Regional load balancer logs, payment reconciliation system

> **Key observation:** The `us-east-1` region was disproportionately affected because it received the configuration change first and absorbed redirected traffic from the other two regions before they were also impacted. This “thundering herd” effect amplified the impact by approximately 40%.

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SECTION 3

Root cause analysis

The incident resulted from the convergence of three failures: a human error in configuration, a gap in automated validation, and insufficient staging fidelity.

01

YAML Indentation Error

The `max_connections_per_node` value was nested under the wrong parent key due to a 2-space indentation error, causing it to be interpreted as 100 (the default) rather than the intended 10,000.

02

Missing Schema Validation

The deployment pipeline validated YAML syntax but not semantic correctness. No range check existed for connection limits — any integer value was accepted without warning.

03

Low-Fidelity Staging

The staging environment processes <50 req/s vs. production's 45,000 req/s. The misconfigured limit of 100 connections was never breached in staging, so the error went undetected.

TECHNICAL DETAIL

The configuration diff that caused the outage

The intended change was to update TLS cipher suites (line 47). However, the engineer's editor auto-formatted adjacent lines, shifting `max_connections_per_node: 10000` from under `connection_pool` to under `rate_limiting`, where it was silently ignored. The `rate_limiting` section already had a default `max_connections` of 100.

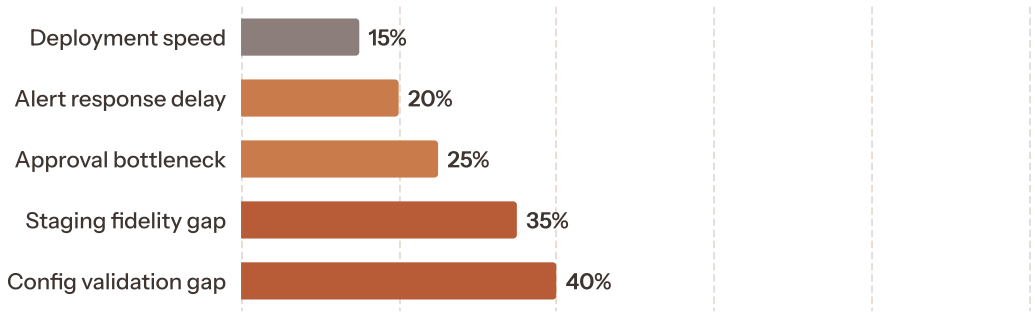
```
# Intended (v3.14.1)
connection_pool:
  max_connections_per_node: 10000
  idle_timeout: 30s
rate_limiting:
  max_requests_per_second: 5000

# Actual (v3.14.2) — note the indentation shift
connection_pool:
  idle_timeout: 30s
rate_limiting:
  max_connections_per_node: 10000  # ignored here
  max_requests_per_second: 5000
  # default max_connections: 100 applied
```

EXHIBIT 4

Contributing factors by category

Weighted assessment of factors that enabled the incident



Source: Post-mortem working group assessment

> *“This wasn’t a failure of one person making a typo. It was a failure of every safety net between that typo and production.”*
>
> — VP of Engineering, post-mortem review meeting

SECTION 4

Remediation actions

Five action items were identified, prioritized by blast radius and implementation complexity. Three have been completed within 72 hours of the incident.

EXHIBIT 5

Action items and completion status

Prioritized remediation plan with owners and deadlines

PRIORITY	ACTION ITEM	OWNER	DEADLINE	STATUS
P0	Add schema validation with range checks to gateway config pipeline	Platform Team	Mar 10	COMPLETE
P0	Implement automatic rollback on error rate threshold breach (>5%)	SRE	Mar 10	COMPLETE
P1	Add synthetic traffic generation to staging (match production load profile)	QA Engineering	Mar 21	IN PROGRESS
P1	Remove dual-approval requirement for emergency rollbacks	SRE	Mar 11	COMPLETE

PRIORITY	ACTION ITEM	OWNER	DEADLINE	STATUS
P2	Implement region-staggered deployments with 15-minute soak between regions	Platform Team	Apr 4	IN PROGRESS

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Validate semantics, not just syntax

Every configuration parameter with operational impact now has defined min/max ranges. Values outside these ranges block deployment and alert the on-call team.

⏳

Automate rollback decisions

If error rate exceeds 5% within 10 minutes of a deployment, the system now automatically reverts to the last known-good configuration without human intervention.

🎯

Test at production scale

Staging environment is being upgraded to replay sampled production traffic at 80% of peak volume. This closes the gap that allowed the misconfiguration to pass validation.

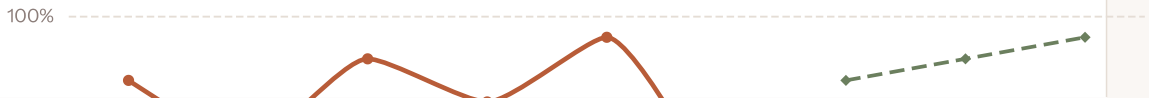
Remove rollback friction

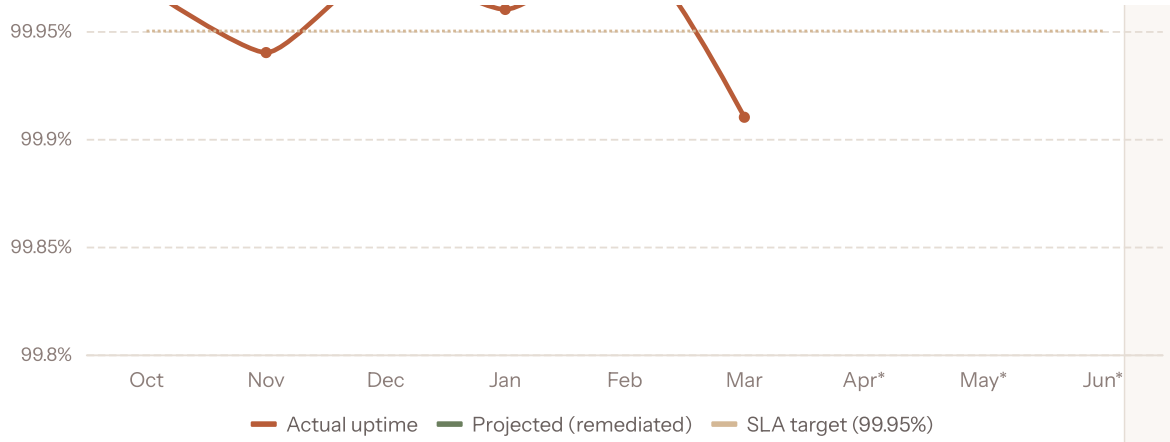
Emergency rollbacks now require single-engineer approval. The 27-minute delay waiting for a second approver during this incident was an unacceptable bottleneck.

EXHIBIT 6

Projected uptime improvement

Estimated monthly uptime with remediation measures in place





Source: SRE modeling based on historical incident frequency and MTTR reduction

SECTION 5

Lessons learned

What went well. The on-call team correctly identified the root cause within 31 minutes of the first alert. Cross-team communication was effective once the incident was escalated to SEV-1 — the war room had representatives from SRE, Platform, and Engineering within 15 minutes. Customer communications were published within 45 minutes of impact confirmation.

The staging gap is a known risk. This is the third incident in 12 months where a configuration or code change passed staging but failed in production due to traffic volume differences. The investment in production-scale staging is overdue.

What went poorly. Initial triage dismissed the first alert as transient, costing approximately 7 minutes of response time. The dual-approval rollback requirement — designed to prevent accidental rollbacks — became a liability during an active incident, adding 27 minutes. The global configuration sync propagated the bad config to healthy regions before the problem was understood.

Blameless culture held. The engineer who authored the configuration change participated in the post-mortem openly. The focus remained on systemic gaps rather than individual error — which is critical, because the real failures were in the safety nets, not the typo.

> *"A two-space indentation error should not be able to take down a global platform. If it can, the problem is the platform, not the spaces."*
>
> — Director of SRE, post-mortem summary

[1] Revenue impact calculated from failed transaction logs, reconciled against payment processor records. Recovered revenue represents transactions successfully retried within 24h.

[2] SLA credits are contractually defined in enterprise agreements and calculated based on monthly downtime minutes exceeding the committed threshold.

[3] Uptime projections assume all P0 and P1 remediation items are fully implemented and operational.

This document is classified as Internal. Distribution is limited to engineering leadership, SRE, and the incident review board. For questions, contact sre-postmortems@arcturus.io.

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